

Section: Electricity

Topic: Semiconductors, Sources, Internal Resistance

A student investigates the characteristics of a light-emitting diode (LED).

(a)

Explain why a resistor is included in series with the LED

✔ The resistor **limits the current** through the LED.  
Without the resistor, the current could rise too high and **damage** the LED — because LEDs have **very low resistance** once conducting.

(b)

The LED has a threshold voltage of 1.8 V.  
Explain what is meant by threshold voltage

✔ The **threshold voltage** is the **minimum voltage** required before the LED **starts to conduct** and emit light.  
Below this voltage, **almost no current** flows through the LED.

(c)

The current through the LED is 15 mA when the potential difference is 2.4 V.  
Calculate the power dissipated in the LED

Use:

$$P = IV = 0.015 \times 2.4 = 0.036 \text{ W or } 36 \text{ mW}$$

(d)

Calculate the resistance of the resistor in series with the LED

Voltage across resistor:

$$V_{\text{resistor}} = 2.4 - 1.8 = 0.6 \text{ V}$$

Use Ohm’s law:

$$R = \frac{V}{I} = \frac{0.6}{0.015} = 40 \, \Omega$$

🧠 Revision Tips

- A **series resistor** protects the LED from excessive current
- Threshold voltage = “**switch-on**” **voltage** for a diode
- Always subtract threshold voltage from total to find voltage across the resistor

- Power in an LED:

$$P = IV$$

where  $V$  is the voltage across the LED